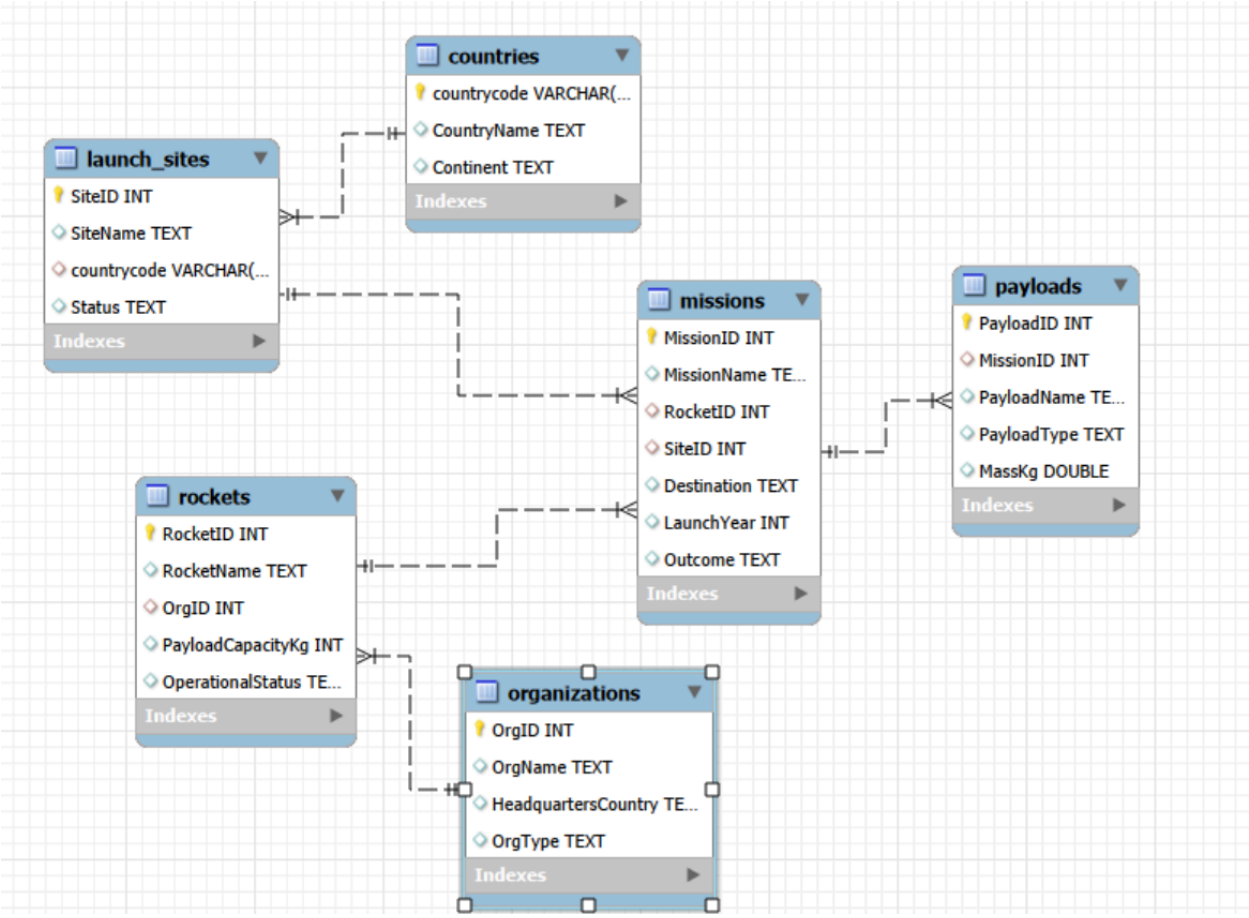
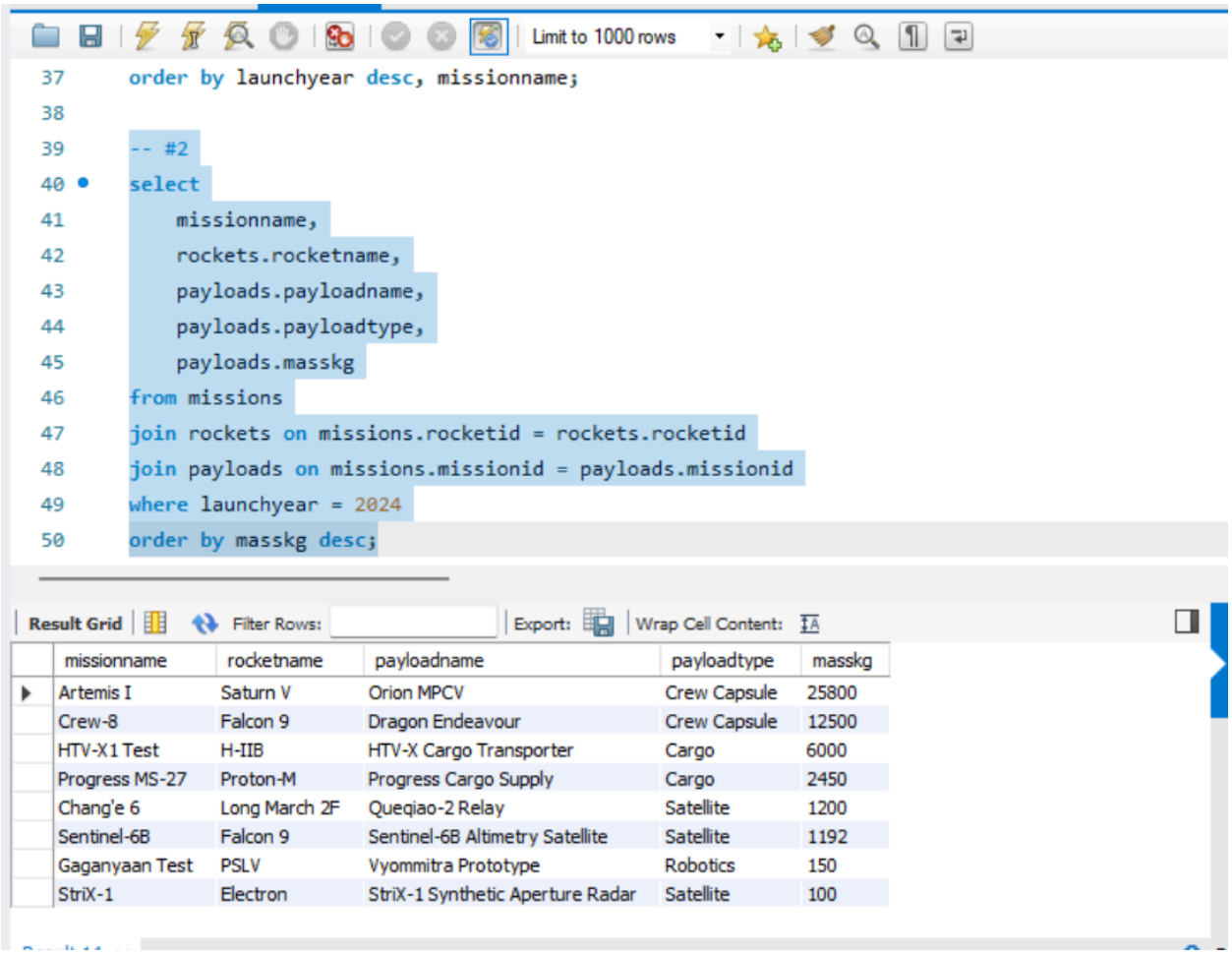
Colin Donahoe
Assignment 4
DBMS



```sql
-- #1
select
    missionname,
    rockets.rocketname,
    organizations.orgname,
    launch_sites.sitename,
    destination,
    launchyear
from missions
join rockets on missions.rocketid = rockets.rocketid
join launch_sites on missions.siteid = launch_sites.siteid
join organizations on rockets.orgid = organizations.orgid
order by launchyear desc, missionname;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

missionname	rocketname	orgname	sitename	destination	lau
Artemis I	Saturn V	National Aeronautics and Space Administration	Kennedy Space Center	Moon	202
Chang'e 6	Long March 2F	China National Space Administration	Jiuquan Satellite Launch Center	Moon	202
Crew-8	Falcon 9	Space Exploration Technologies Corp	Kennedy Space Center	ISS	202
Gaganyaan Test	PSLV	Indian Space Research Organisation	Satish Dhawan Space Centre	LEO	202
HTV-X1 Test	H-IIB	Japan Aerospace Exploration Agency	Tanegashima Space Center	ISS	202
Progress MS-27	Proton-M	Roscosmos State Corporation	Baikonur Cosmodrome	ISS	202
Sentinel-6B	Falcon 9	Space Exploration Technologies Corp	Vandenberg Space Force Base	LEO	202
StriX-1	Electron	Rocket Lab	Mahia Launch Complex 1	LEO	202



```
37    order by launchyear desc, missionname;
38
39    -- #2
40    select
41        missionname,
42        rockets.rocketname,
43        payloads.payloadname,
44        payloads.payloadtype,
45        payloads.masskg
46    from missions
47    join rockets on missions.rocketid = rockets.rocketid
48    join payloads on missions.missionid = payloads.missionid
49    where launchyear = 2024
50    order by masskg desc;
```

Limit to 1000 rows

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

missionname	rocketname	payloadname	payloadtype	masskg
Artemis I	Saturn V	Orion MPCV	Crew Capsule	25800
Crew-8	Falcon 9	Dragon Endeavour	Crew Capsule	12500
HTV-X1 Test	H-IIB	HTV-X Cargo Transporter	Cargo	6000
Progress MS-27	Proton-M	Progress Cargo Supply	Cargo	2450
Chang'e 6	Long March 2F	Queqiao-2 Relay	Satellite	1200
Sentinel-6B	Falcon 9	Sentinel-6B Altimetry Satellite	Satellite	1192
Gaganyaan Test	PSLV	Vyommitra Prototype	Robotics	150
StriX-1	Electron	StriX-1 Synthetic Aperture Radar	Satellite	100

```sql
-- #3
select
    sitename,
    missions.missionname,
    rockets.rocketname,
    missions.launchyear
from launch_sites
left join missions on launch_sites.siteid = missions.siteid
left join rockets on missions.rocketid = rockets.rocketid
where launchyear is null or launchyear = 2024
order by sitename;

 -- #4
with payload_sums as(
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

sitename	missionname	rocketname	launchyear
Baikonur Cosmodrome	Progress MS-27	Proton-M	2024
Guiana Space Centre	NULL	NULL	NULL
Jiuquan Satellite Launch Center	Chang'e 6	Long March 2F	2024
Kennedy Space Center	Crew-8	Falcon 9	2024
Kennedy Space Center	Artemis I	Saturn V	2024
Mahia Launch Complex 1	StriX-1	Electron	2024
Satish Dhawan Space Centre	Gaganyaan Test	PSLV	2024
Tanegashima Space Center	HTV-X1 Test	H-IIB	2024
Vandenberg Space Force Base	Sentinel-6B	Falcon 9	2024

Result 15

HW1* HW3 hw4*

Limit to 1000 rows

```sql
with payload_sums as(
    select
        organizations.orgid,
        sum(payloads.masskg) as total_payload_mass
    from organizations
    join rockets on rockets.orgid = organizations.orgid
    join missions on missions.rocketid = rockets.rocketid
    join payloads on payloads.missionid = missions.missionid
    group by organizations.orgid
) select
    orgname,
    headquarterscountry,
    orgtype,
    total_payload_mass
from organizations
join payload_sums on organizations.orgid = payload_sums.orgid
order by total_payload_mass desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

orgname	headquarterscountry	orgtype	total_payload_mass
National Aeronautics and Space Administration	USA	Government	25800
Space Exploration Technologies Corp	USA	Private	13692
Japan Aerospace Exploration Agency	JPN	Government	6000
Roscosmos State Corporation	RUS	Government	2450
China National Space Administration	CHN	Government	1200

```sql
-- #5
with country_payload_sums as(
    select
        countries.countrycode,
        sum(payloads.masskg) as total_payload_mass
    from countries
    join launch_sites on launch_sites.countrycode = countries.countrycode
    join missions on missions.siteid = launch_sites.siteid
    join payloads on payloads.missionid = missions.missionid
    group by countries.countrycode
),
total_missions as (
    select
        countries.countrycode,
        count(missions.missionid) as total_missions
    from countries
    join launch_sites on launch_sites.countrycode = countries.countrycode
```

```
93      ),
94      total_missions as (
95          select
96              countries.countrycode,
97              count(missions.missionid) as total_missions
98          from countries
99          join launch_sites on launch_sites.countrycode = countries.countrycode
100         join missions on missions.siteid = launch_sites.siteid
101         group by countries.countrycode
102     ) select
103         countryname,
104         total_missions,
105         total_payload_mass
106     from countries
107     join country_payload_sums on country_payload_sums.countrycode = countries.countrycode
108     join total_missions on total_missions.countrycode = countries.countrycode
109     order by total_payload_mass desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

countryname	total_missions	total_payload_mass
United States	3	39492
Japan	1	6000
Russia	1	2450
China	1	1200
India	1	150